deviceDiscovery

Service Description

Service ID: *"deviceDiscovery"*

Abstract
This document describes the `deviceDiscovery` service, version 1.0.0, produced by the ServiceRegistry core system of Arrowhead 5.2.2.

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1 Overview

This document describes the `deviceDiscovery` service, which provides registration and lookup of the devices of the local cloud: the physical or virtual hosts, with their network addresses and metadata, on which the systems run. It is produced by the ServiceRegistry system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interfaces are specified in the HTTP/TLS/JSON IDD [2] and the MQTT/TLS/JSON IDD [3].

1.1 Significant Prior Art

Arrowhead 4.x had no device concept in the ServiceRegistry; the separate DeviceRegistry and SystemRegistry onboarding systems of 4.4 held devices. In 5.x the device is part of the ServiceRegistry model and is optional for a system.

1.2 How This Service Is Meant to Be Used

A device (or the onboarding procedure on its behalf) registers itself with `register` and later removes its entry with `revoke`. Other systems use `lookup` to find devices by name, address or metadata, e.g. to know which systems share a host.

1.3 Important Delimitations and Dependencies

The service does not prove that a device exists or owns the addresses it registers; that is the job of the onboarding procedure and the device certificate. A device with systems assigned cannot be revoked (423 Locked). Address validation follows the `allow.self.addressing` and `allow.non.routable.addressing` settings of the ServiceRegistry.

2 Service Interface

This section describes the interfaces to the `deviceDiscovery` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.



Figure 1: SysML-style block of the `deviceDiscovery` service and its operations.

The following interface operations are available.

2.1 operation register (DeviceRequest) : DeviceResponse

Registers a device with its addresses and metadata. If a device with the same name and the same data exists, it is returned unchanged (200); a new device is created (201). A device with the same name and different data is refused.

2.2 operation lookup (DeviceLookupRequest) : DeviceListResponse

Returns the devices matching all given filters; without filters all devices.

2.3 operation revoke (Name)

Removes the named device. Returns 200 if it was removed, 204 if it did not exist, and 423 Locked if systems are still assigned to it.

3 Information Model

Here, all data objects that can be part of the `deviceDiscovery` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.7, which are used to represent things like hashes and identifiers.

3.1 struct Address

A network address with its type.

Field	Type	Description
type	String	Address type: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
address	String	One network address (host name, IPv4, IPv6 or MAC address).

3.2 struct DeviceListResponse

List of devices.

Field	Type	Description
entries	List<DeviceResponse>	Matching records of the current page.
count	Number	Total number of matching records (all pages).

3.3 struct DeviceLookupRequest

Filter for devices; all fields optional, combined with AND.

Field	Type	Description
deviceNames	List<Name>	Device names to match.
addresses	List<String>	Network addresses of the entity.
addressType	String	Address type filter: `HOSTNAME`, `IPV4`, `IPV6` or `MAC`.
metadataRequirementList	List<MetadataRequirement>	Metadata requirements; each map is one alternative.

3.4 struct DeviceRequest

A device and its addresses.

Field	Type	Description
name	Name	Device name, PascalCase.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<String>	Network addresses of the entity.

3.5 struct DeviceResponse

A registered device.

Field	Type	Description
name	Name	Device name.
metadata	Metadata	Free-form metadata (JSON object).
addresses	List<Address>	Network addresses with their types.
createdAt	DateTime	UTC time of creation.
updatedAt	DateTime	UTC time of the last change.

3.6 struct MetadataRequirement

A JSON object of metadata requirements: each key is a metadata path (dot notation), each value the required value or an operator object such as `{"op": "LESS_THAN", "value": 10}`.

3.7 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
List	An *array* of a known number of items, each having type A.
Metadata	A set of key-value pairs with arbitrary, possibly nested values.
Name	A string identifier that is intended to be both human and machine-readable.
Number	An integer number.
String	A chain of characters.

4 References

[1] J. Delsing, “ServiceRegistry – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, serviceregistry/documentation/v5.2.2/ServiceRegistry-v5.2.2-SysD.pdf.

[2] ——, “deviceDiscovery HTTP/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.2.2, 2026, serviceRegistry-v5.2.2-deviceDiscovery-HTTP-IDD.pdf.

[3] ——, “deviceDiscovery MQTT/TLS/JSON – Interface Design Description (ServiceRegistry),” Eclipse Arrowhead, IDD 5.2.2, 2026, serviceRegistry-v5.2.2-deviceDiscovery-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	